

AgentSkills '26: The First Workshop on Agent Skills — Design, Evaluation, and Optimization of Procedural Knowledge for LLM Agents

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Abstract

Agent Skills are structured packages of procedural knowledge that augment LLM agents at inference time without model modification. Their rapid adoption across agent platforms has created urgent questions about authoring, evaluation, synthesis, composition, and security. AgentSkills '26 will bring together researchers and practitioners working on Skills benchmarking, optimization, domain applications, and ecosystem infrastructure. The workshop will feature invited talks, peer-reviewed papers, a live Skills design challenge, and a panel discussion.

Keywords: agent skills, LLM agents, procedural knowledge, agent augmentation, benchmarking, agentic systems

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1 Topic Summary and Relevance

Large language models have evolved from text generators into autonomous agents capable of executing complex, multi-step tasks in real-world environments [1]. However, a fundamental tension persists: foundation models provide broad capabilities but lack the procedural knowledge required for domain-specific workflows, while fine-tuning is expensive and sacrifices generality.

Agent Skills offer an emerging solution. A Skill is a structured package comprising a SKILL.md file with procedural instructions plus optional executable resources that augment agent behavior at inference time without model modification [2]. Skills have rapidly become a cross-platform standard: Anthropic introduced Skills for Claude Code; Google adopted the same format for Gemini CLI; Microsoft and Vercel maintain curated Skills repositories. OpenClaw, the largest open-source agent framework, treats Skills as a first-class extension interface, and ClawHub provides a public registry with publishing, versioning, search, and moderation

for SKILL.md packages [13, 14]. This convergence across competing platforms signals that Skills address a genuine architectural need.

A converging research area. Independent research threads are converging on Skills as a first-class object of study. Voyager [7] introduced skill libraries for embodied agents; SSO [8] formalized skill composition at ICML 2024; GEPA [6] demonstrated fully automatic Skill synthesis (ICLR 2026 Oral). SkillsBench [5] provided the first benchmark treating Skills as first-class artifacts, revealing that curated Skills improve agent pass rates by 16.2 percentage points on average, yet self-generated Skills provide negligible benefit and can *hurt* performance in nearly a third of tasks. Meanwhile, the rapid growth of Skills ecosystems has outpaced security practices: an audit of 3,984 Skills found that 37% contained at least one security flaw [19], and recent attack studies demonstrate both supply-chain infiltration and prompt injection via malicious Skills [16, 18]. Perhaps the clearest sign of momentum is temporal concentration: after isolated precursors in 2023–2025 [7–9], at least eight focused Skills papers appeared between Feb. 8 and Mar. 3, 2026 alone [3–5, 10, 11, 15–17].

Relevance to CAIS. Agent Skills sit at the intersection of core CAIS themes: agentic system design, evaluation methodology, and human–AI collaboration. The workshop is uniquely positioned at CAIS because Skills span the full stack—from theoretical foundations (composition, optimization) to systems engineering (ecosystems, security) to applied deployment—and no existing venue addresses this intersection.

1.1 Workshop Scope and Topics

We invite contributions on topics including:

- **Skills design and authoring:** Principles for writing effective Skills; structure–efficacy relationships; domain-specific design patterns; failure mode analysis.
- **Skills evaluation and benchmarking:** Evaluation frameworks treating Skills as first-class artifacts; paired evaluation protocols; metrics beyond pass rate (cost, token efficiency, reliability).
- **Automatic Skills synthesis and optimization:** LLM-based Skills generation (e.g., GEPA [6]); reinforcement learning for Skills refinement; prompt optimization for procedural guidance; evolutionary approaches.
- **Skills composition and scaling:** Compositional effects of multiple Skills; interference and synergy patterns; optimal Skills selection and retrieval for novel tasks.
- **Domain-specific Skills:** Healthcare, manufacturing, scientific computing, cybersecurity, and other vertical applications; domain expert–AI co-authoring workflows.
- **Skills ecosystems and infrastructure:** Quality assurance and verification; registries, versioning, and distribution; interoperability across agent harnesses.
- **Skills security and supply chain:** Malicious Skill detection; formal verification of Skill behavior; sandboxing and permission models; supply-chain attacks and defenses for

Skill registries.

- **Continual learning with Skills:** Feedback-based Skills improvement; Skills as a medium for agent self-improvement; trajectory mining for Skills discovery.

2 Previous Iterations

This is the **inaugural** edition of AgentSkills. While no prior edition exists, the field has reached a critical mass that demands a dedicated venue. SkillsBench [5] received 4 citations within 14 days of release, and the organizers have run a Skilathon with 41+ participants. The SkillsBench ecosystem analysis highlights the pace: after deduplication it retained 47,150 unique Skills from 6,323 repositories, and daily additions peaked at 18,904 in January 2026 [5]. Recent February–March 2026 work spans marketplace-scale measurement [3], industrial deployment with smaller models [4], attack benchmarking [16], automated Skill discovery [10], large-scale Skill registries [12], and ecosystem-level orchestration [11], suggesting that the area is moving from isolated prototypes toward a sustained research and infrastructure stack. The organizing team has experience running workshops and tutorials at ICML, NeurIPS, and ICLR.

3 Workshop Structure and Format

The workshop will be a **full-day, in-person event** designed to maximize discussion and community building. Confirmed invited speaker: Dawn Song (UC Berkeley). Tentative invited speakers and panelists include Alex Shaw (Laude), Manling Li (Northwestern University and Amazon Scholar), and Qingsong Wen (Squirrel AI Learning). The live design challenge is meant to make the event action-oriented rather than purely presentational: participants will be given a real-world task from SkillsBench, author a Skill under a time constraint, and have it scored automatically by the benchmark’s deterministic verifiers. The panel will bring together academic researchers, industry practitioners, and platform developers to discuss open problems such as Skill composability, security certification, and standardization across agent harnesses.

- **Morning (9:00–12:30):** opening remarks, keynote on foundations of Agent Skills, coffee break, keynote on benchmarking agent augmentation, four contributed talks, and lightning talks.
- **Lunch (12:30–14:00):** Networking lunch with poster setup
- **Afternoon (14:00–17:30):** keynote on Skills optimization and RL, three contributed talks, poster/demo session, a live Skills design challenge using SkillsBench infrastructure, a panel on the future of procedural knowledge for agents, and closing remarks with a best paper award.

4 Paper Selection and Participant Expectations

We will accept two tracks: *full papers* (up to 6 pages + references) and *short papers / demos* (up to 4 pages + references).

Submissions must follow the ACM SIGPLAN template and will be handled through HotCRP. Each submission will receive at least two single-blind reviews, with decisions based on technical quality, novelty, relevance, and discussion value. We will encourage authors to release Skills, prompts, evaluation harnesses, or other reusable artifacts when appropriate so that the workshop produces community resources in addition to presentations. We expect 30–50 submissions, plan to accept 8–12 papers, and require at least one author of each accepted paper to register, attend, and present.

5 Tentative Schedule and Key Dates

- Website and call for papers: Mar 28, 2026
- Paper submission deadline: Apr 12, 2026 (AoE)
- Author notification: May 5, 2026
- Camera-ready deadline: May 16, 2026 (AoE)
- Workshop day at CAIS '26: May 26, 2026

6 Diversity and Inclusion

Our 14 organizers span industry and academia across 8 institutions and include researchers from security, systems, optimization, and NLP backgrounds. The workshop scope intentionally reaches beyond core ML to communities in healthcare, manufacturing, cybersecurity, and scientific computing. The live Skills design challenge is accessible to participants with different levels of experience, and we will seek student travel grants to support attendance by junior researchers. All accepted papers, challenge artifacts, and invited talk slides will be made publicly available. We will follow the ACM Policy Against Harassment and the CAIS code of conduct.

7 Program Committee (Confirmed)

Xiangyi Li, (BenchFlow), Kyoung Whan Choe (RLWRD), Yifeng He (UC Davis), Songlin Li (Stanford), Tianming Sha (FSU), Lincan Li (FSU), Cheng Cheng (Google), Tiankai Yang (USC), Shawn Li (USC), Yuehan Qin (USC), Jiate Li (USC), Xiyang Hu (ASU), Hanwen Xing (USC), and Ji Ping (Hunter & GC CUNY).

8 Organizer Bios

Xiangyi Li is founder of BenchFlow, lead author of SkillsBench. BenchFlow has been early and innovative in the AI evaluation field. SkillsBench is the mostly mentioned artifact in Agent Skills research. Contact: xiangyi@benchflow.ai.

Wenbo Chen is an Applied Scientist at Amazon working on machine learning and optimization, and a co-first-author of SkillsBench. Contact: wenbo.chen@gatech.edu.

Xuandong Zhao is a postdoctoral researcher at UC Berkeley whose work focuses on LLMs, agentic AI, and AI safety; he contributes to benchmarks including Humanity Last Exam, Terminal Bench, and SkillsBench. Contact: xuandongzhao@berkeley.edu.

Xiaokun Chen is a researcher at Stanford University working on ML systems and efficient LLM inference, and a co-author of SkillsBench. Contact: xiaokun@stanford.edu.

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Yifeng He is a PhD candidate at UC Davis whose research focuses on AI-driven software security and engineering; he is also a maintainer of SkillsBench. Contact: yfhe@ucdavis.edu.

Hao Chen is a professor at UC Davis whose research spans computer security and AI security, including the security of LLM-based agents. Contact: chen@ucdavis.edu.

Yushun Dong is an Assistant Professor at Florida State University working on trustworthy AI, including security, privacy, explainability, and LLM agents. Contact: y24f@fsu.edu.

Yan Liu is a Professor of Computer Science at USC and Director of the USC Machine Learning Center, with research in machine learning for health care, sustainability, and social media analysis. Contact: yanliu.cs@usc.edu.

Yue Zhao is an Assistant Professor at University of Southern California. He is the contributor or creator of widely used open-source machine learning libraries such as PyOD, PyGOD, Agent-Audit, LangSkills, and TrustLLM, which together have accumulated more than 30 million downloads and over 22,000 GitHub stars. Contact: yue.z@usc.edu.

Emilio Ferrara is a Professor of Computer Science and Associate Chair at USC, and the inaugural (ad interim) Director of the Interdisciplinary Data Science Program at the USC School of Advanced Computing. He holds joint appointments in Communication and Preventive Medicine. His research uses AI, data science, and network science to study human behavior in techno-social systems and information networks. He has published over 200 papers and received honors including the AAAI Fellow, DARPA Young Faculty Award, DARPA Director's Fellowship, and the Complex Systems Society Junior Scientific Award. Contact: emiliofe@usc.edu.

Dawn Song is a Professor of Computer Science at UC Berkeley and Co-Director of the Berkeley Center for Responsible Decentralized Intelligence. Her research spans AI safety and security, agentic AI, privacy, and decentralization, and she is an ACM Fellow, IEEE Fellow, MacArthur Fellow, and member of the American Academy of Arts and Sciences. Profile: <https://dawnsong.io/bio.html>.

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